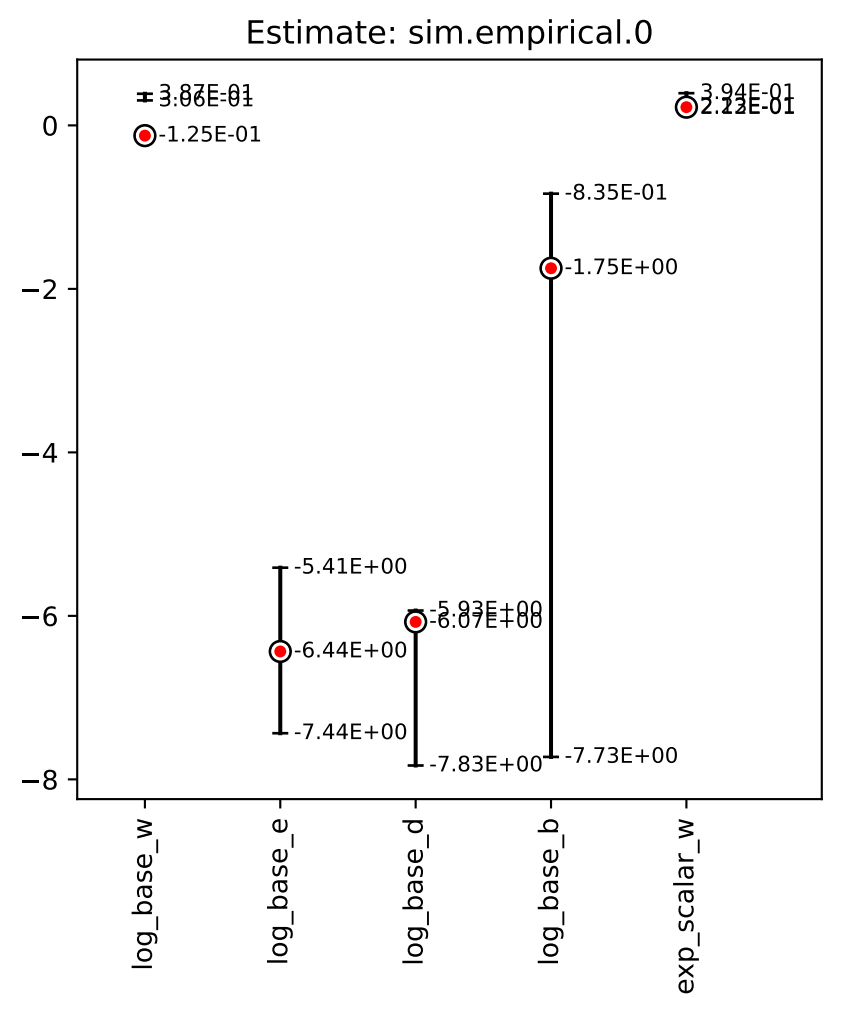
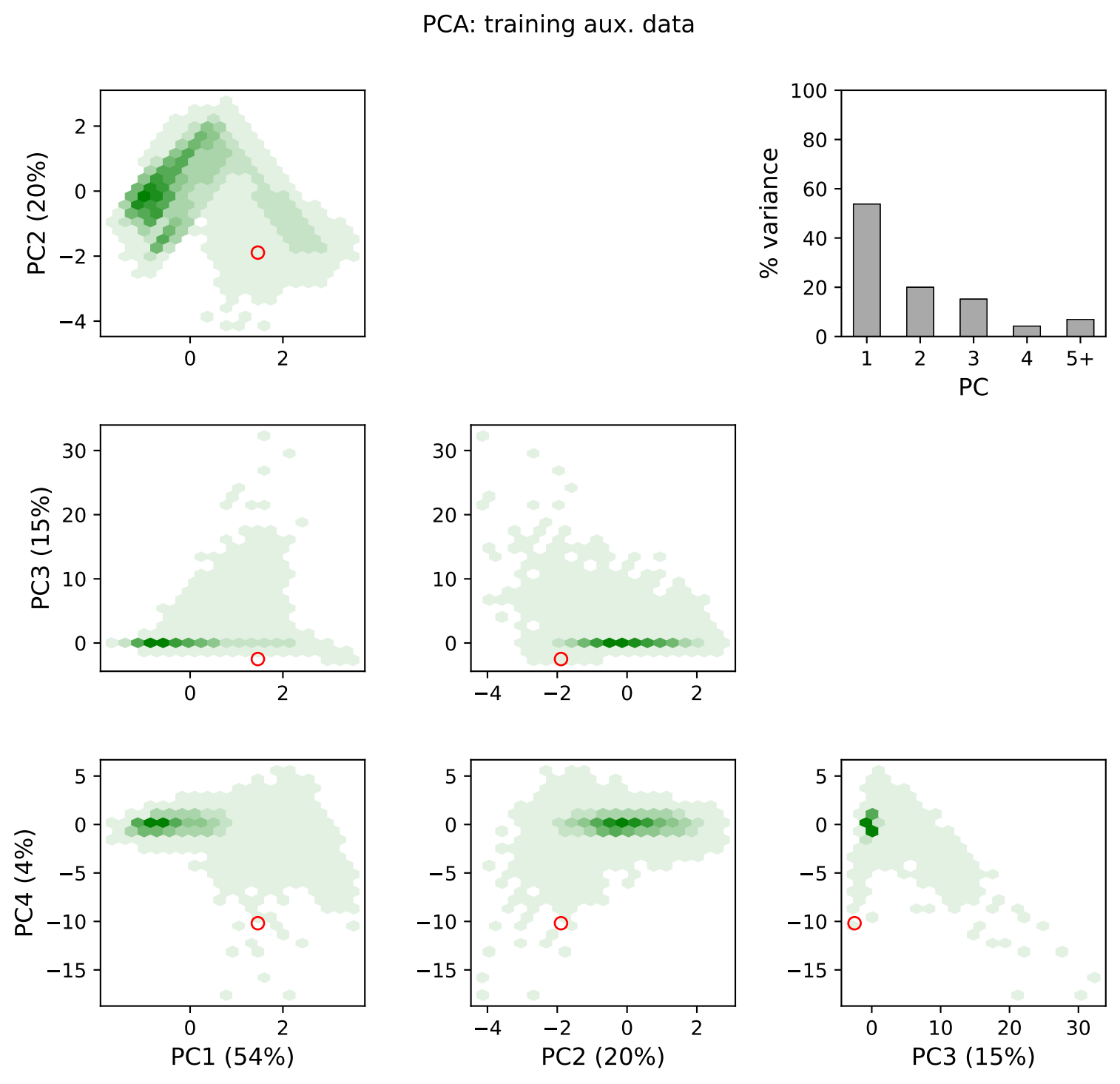
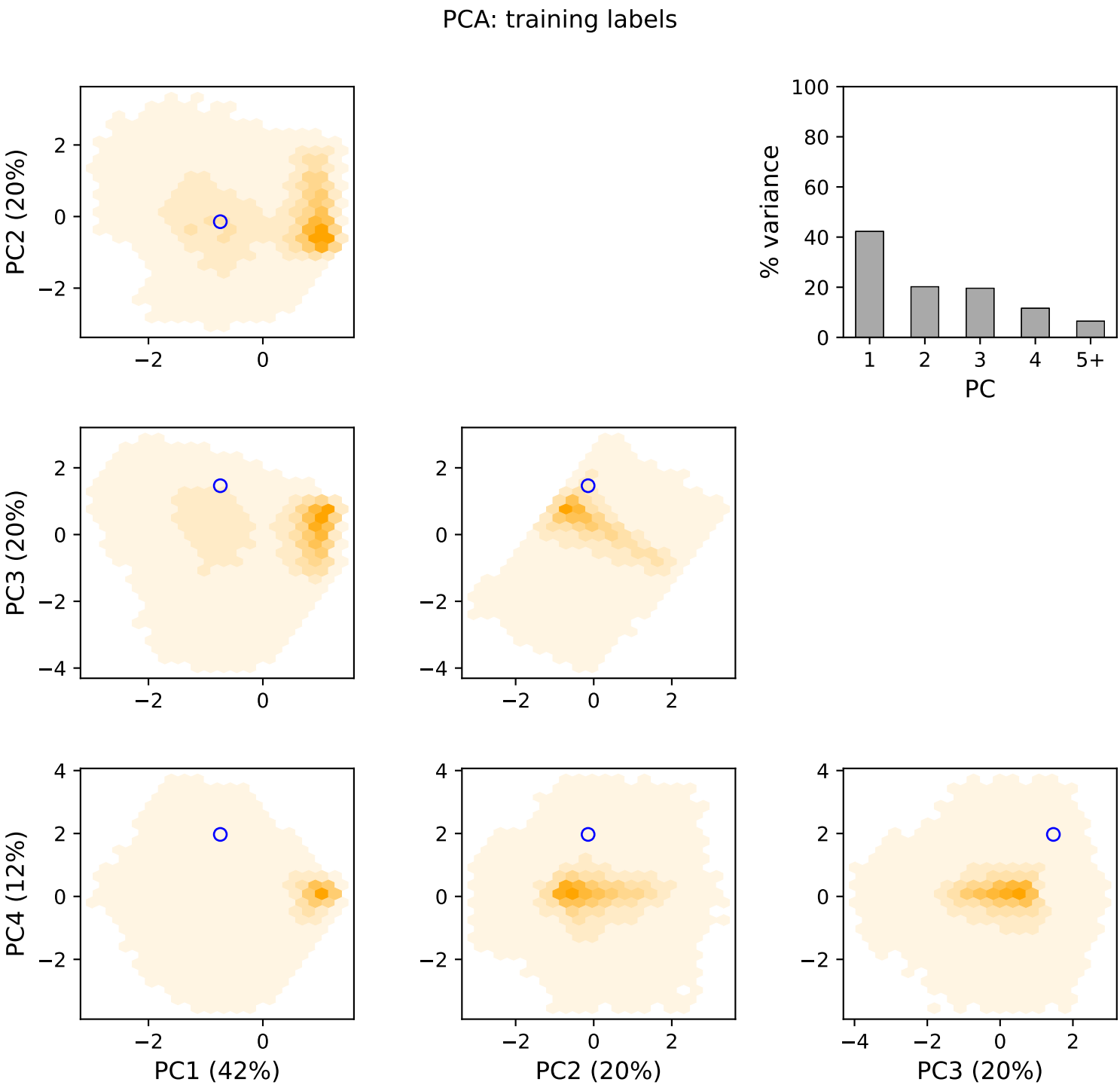
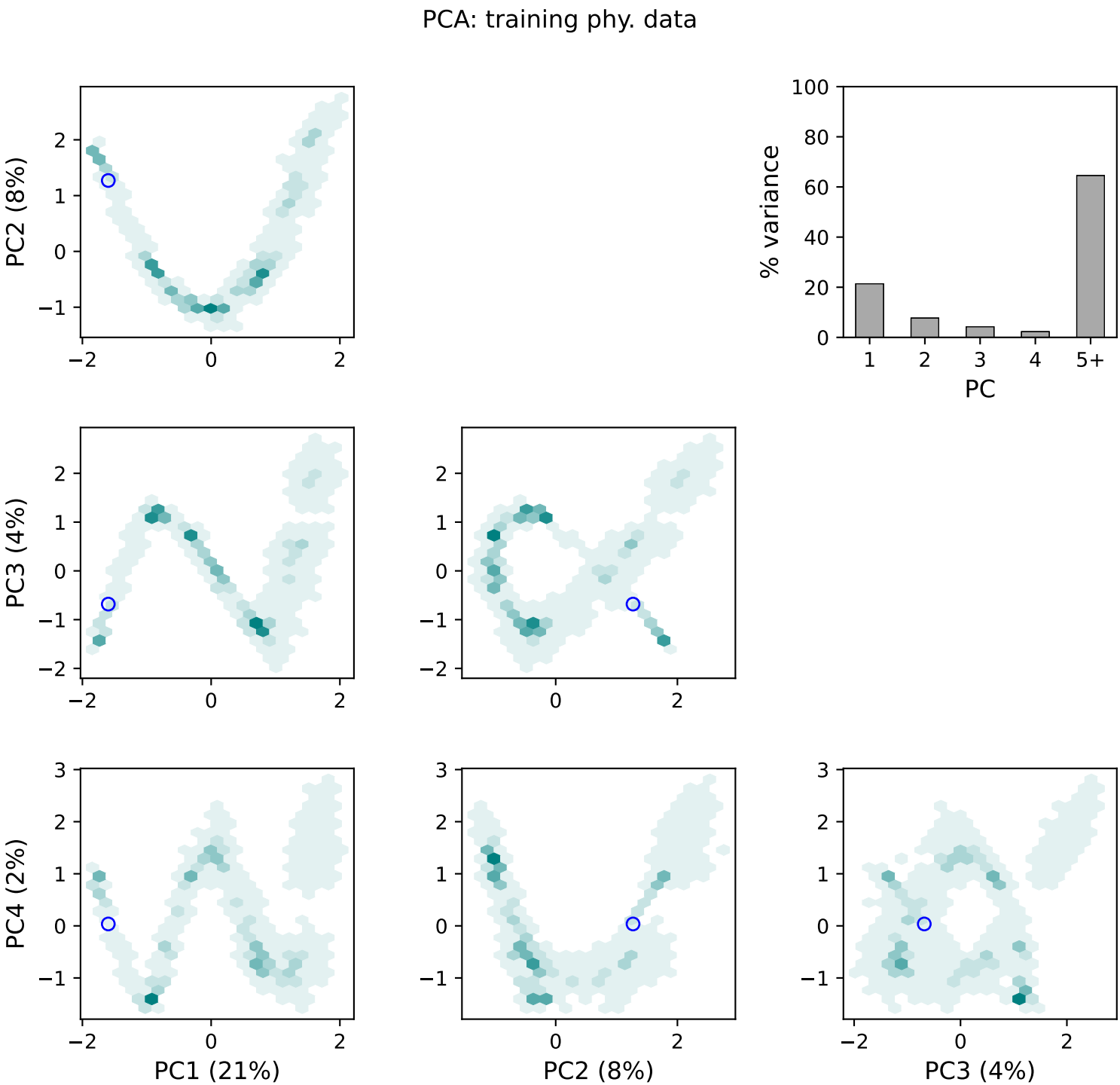
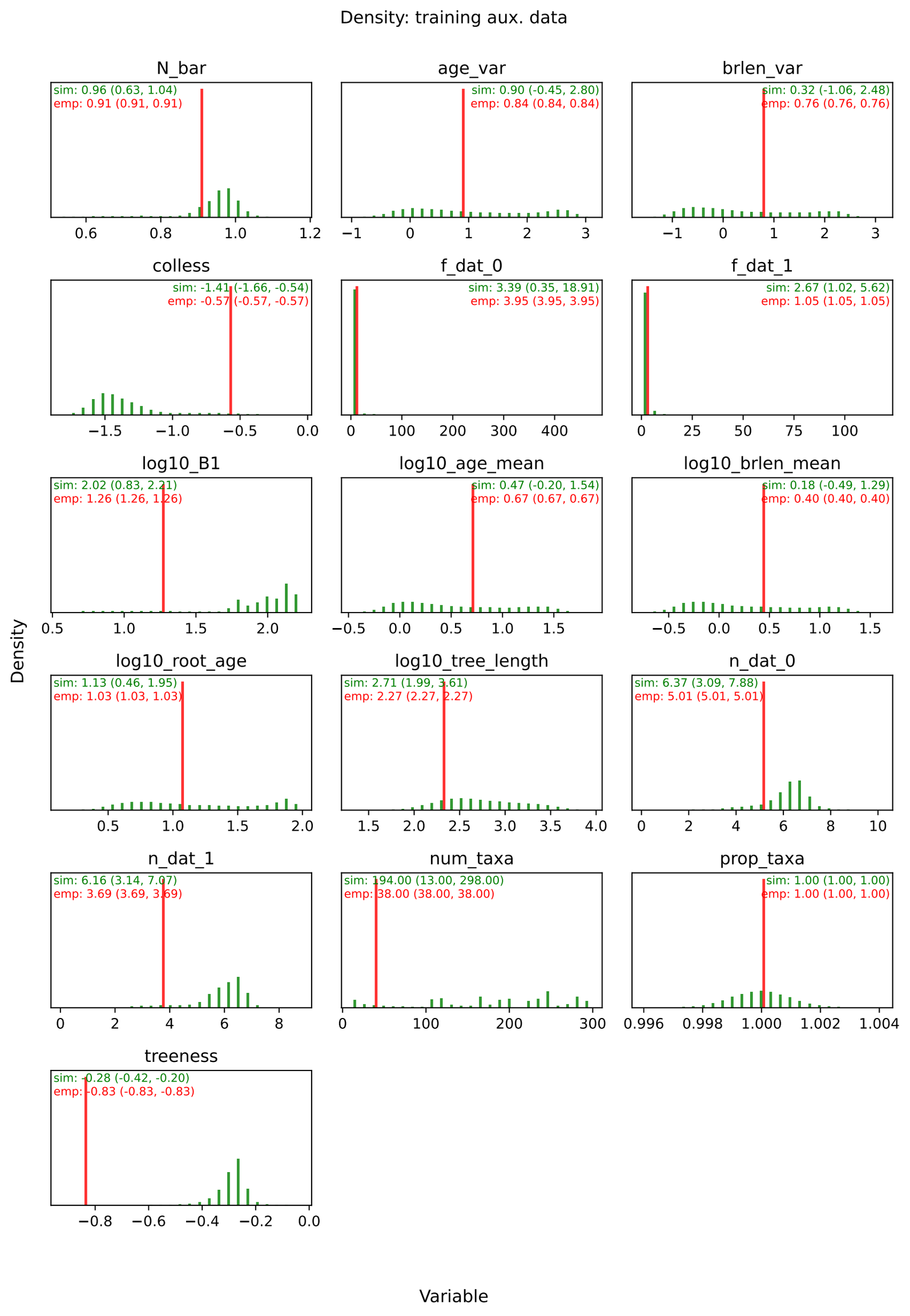
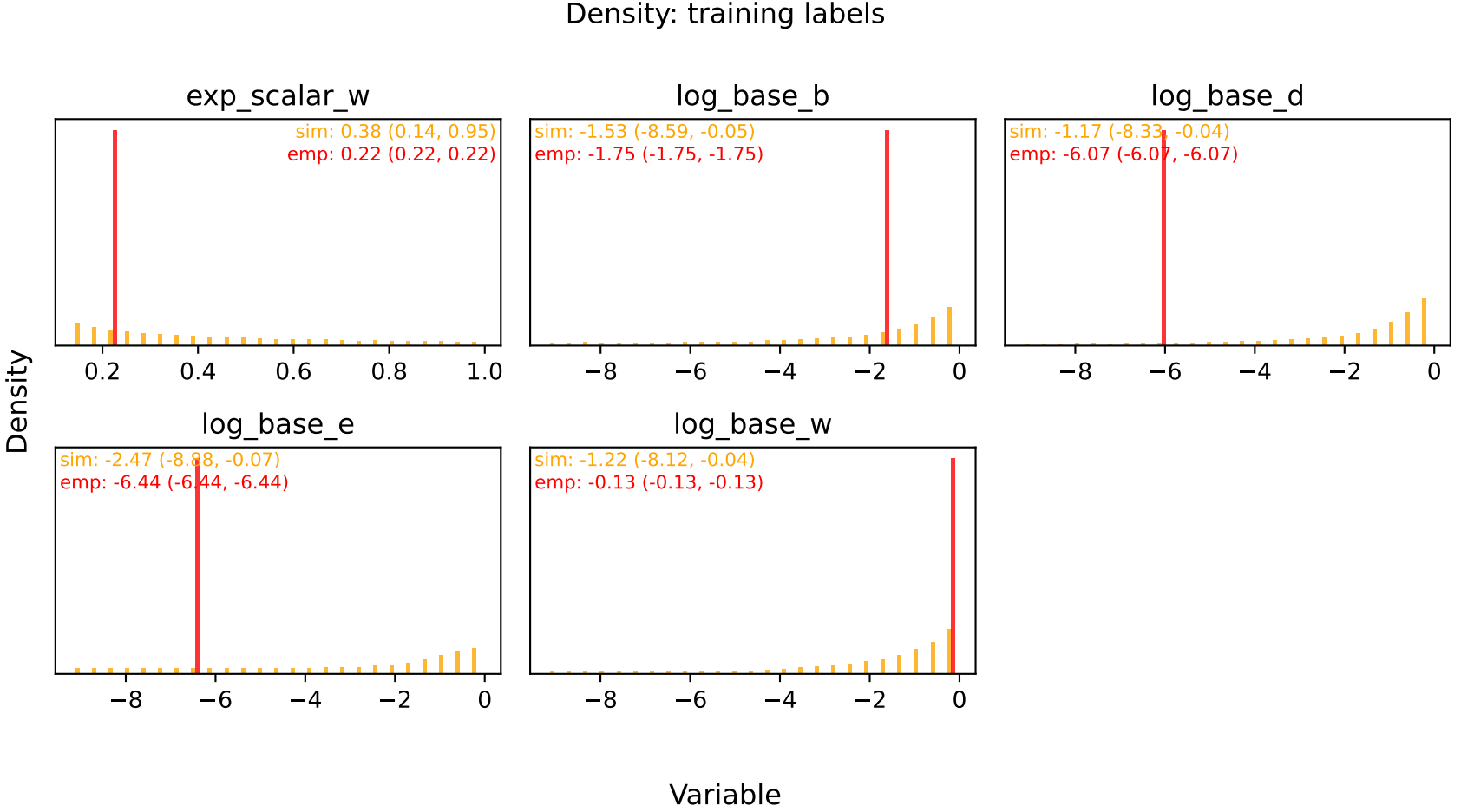

Estimate: sim.empirical.0
3.87E-01
3.06E-01
-1.25E-01
3.94E-01
2.22E-01
-8.35E-01
-1.75E+00
-5.41E+00
-5.93E+00
-6.07E+00
-6.44E+00
-7.44E+00
-7.83E+00
-7.73E+00
0
−2
−4
−6
−8
log_base_w
log_base_e
log_base_d
log_base_b
exp_scalar_w

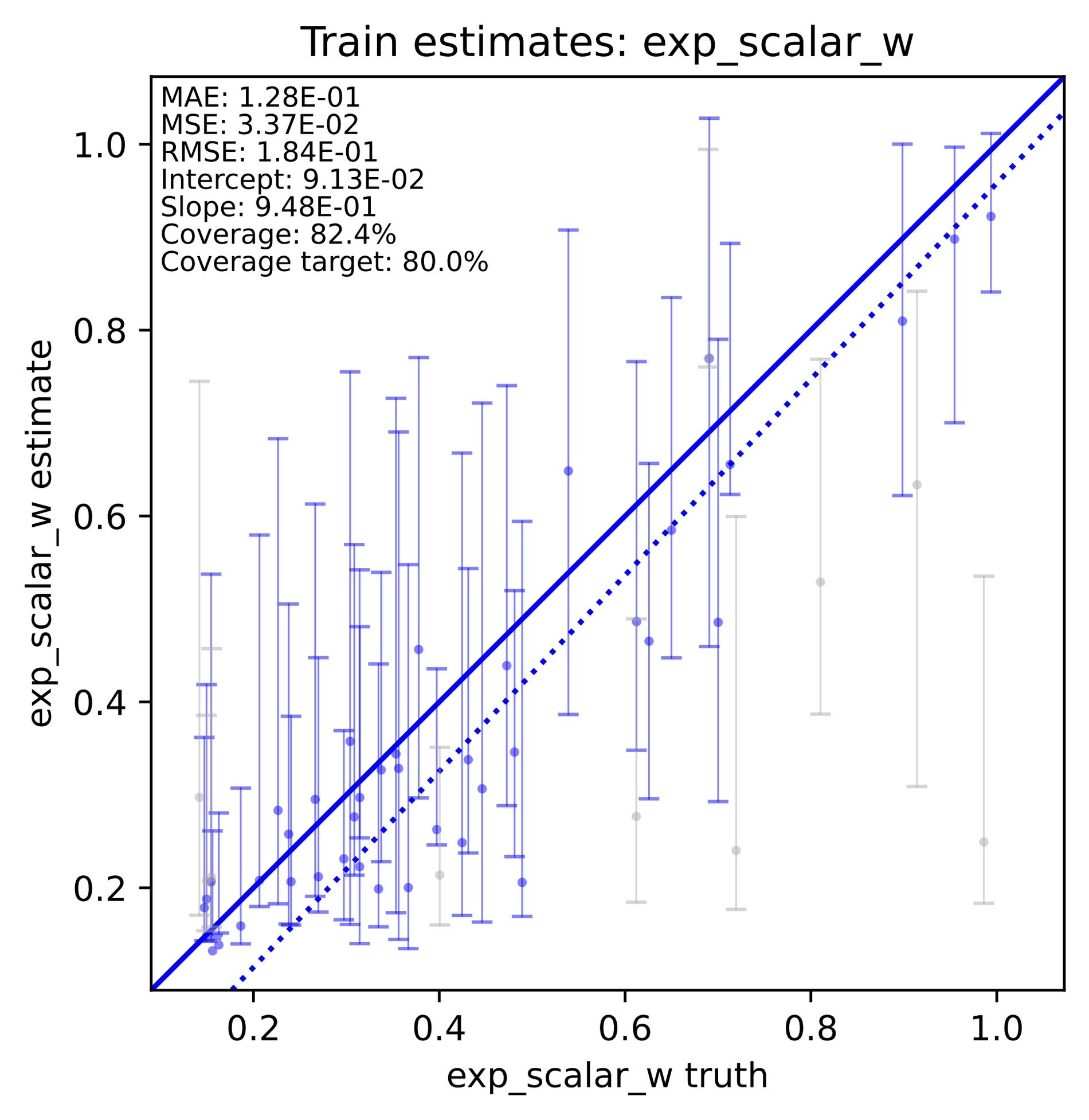

PCA: training aux. data
PC2 (20%)
% variance
PC
PC3 (15%)
PC4 (4%)
PC1 (54%)
PC2 (20%)
PC3 (15%)

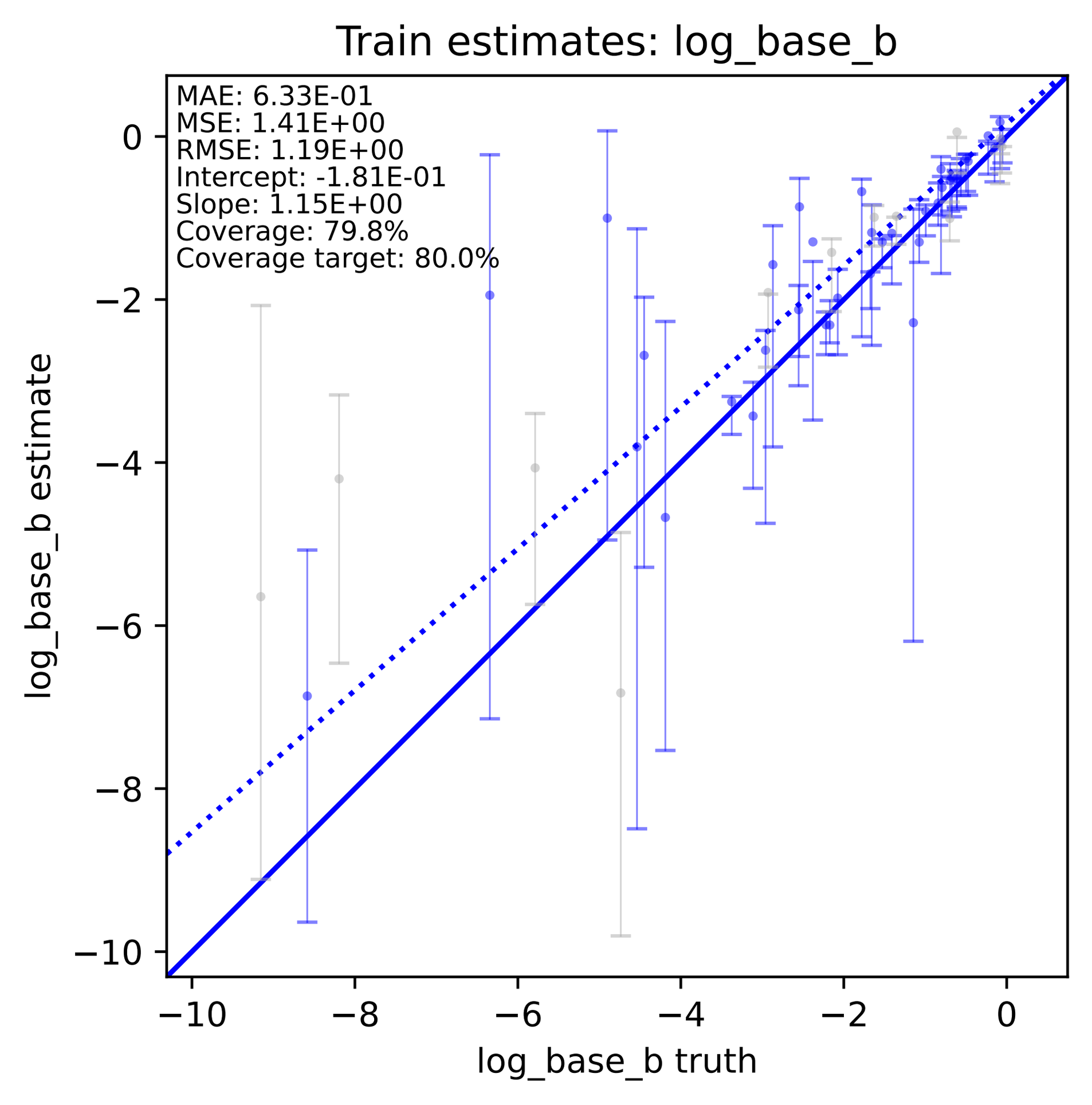

PCA: training labels
PC2 (20%)
% variance
PC
PC3 (20%)
PC4 (12%)
PC1 (42%)
PC2 (20%)
PC3 (20%)

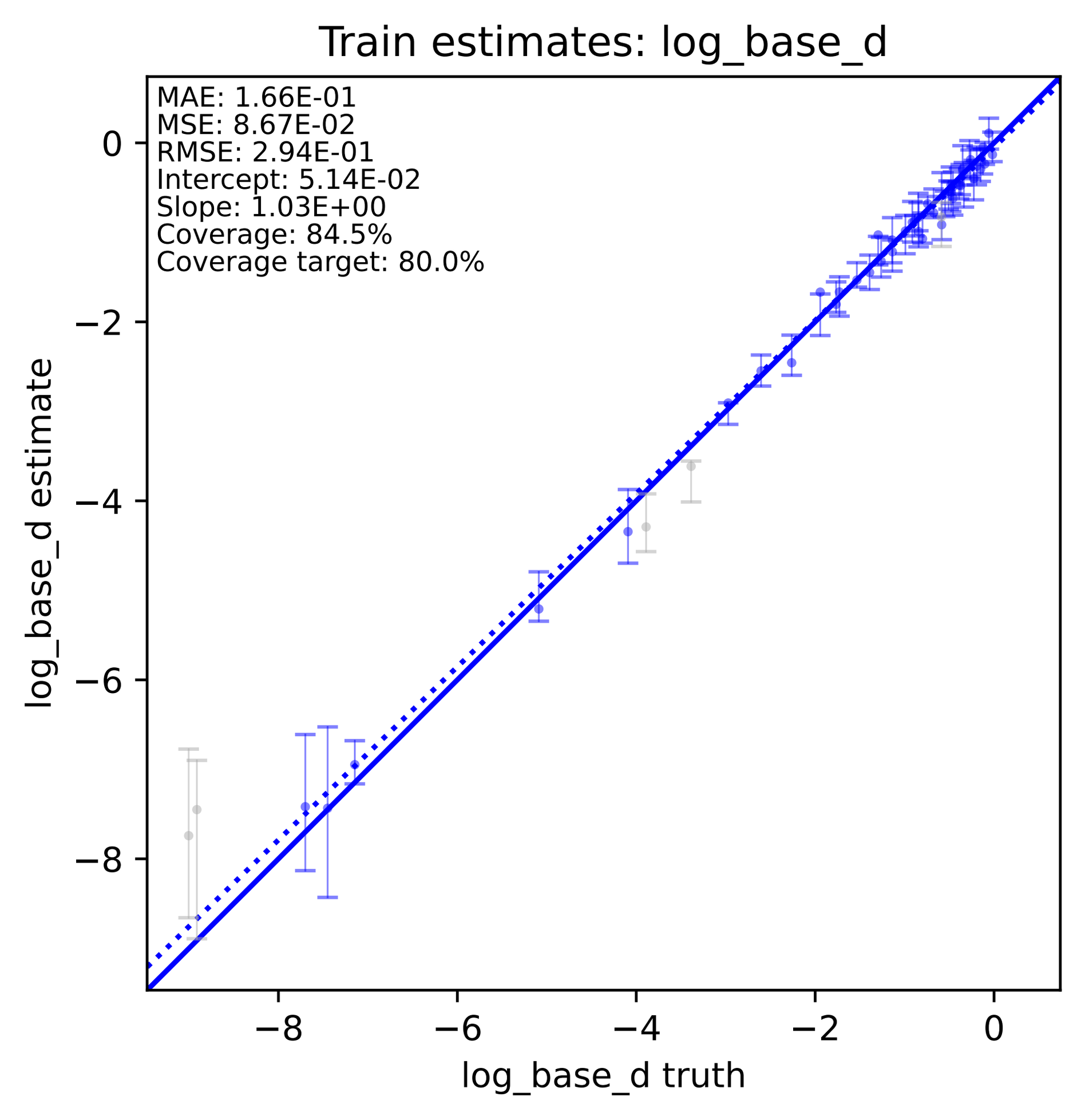

PCA: training phy. data
PC2 (8%)
% variance
PC
PC3 (4%)
PC4 (2%)
PC1 (21%)
PC2 (8%)
PC3 (4%)

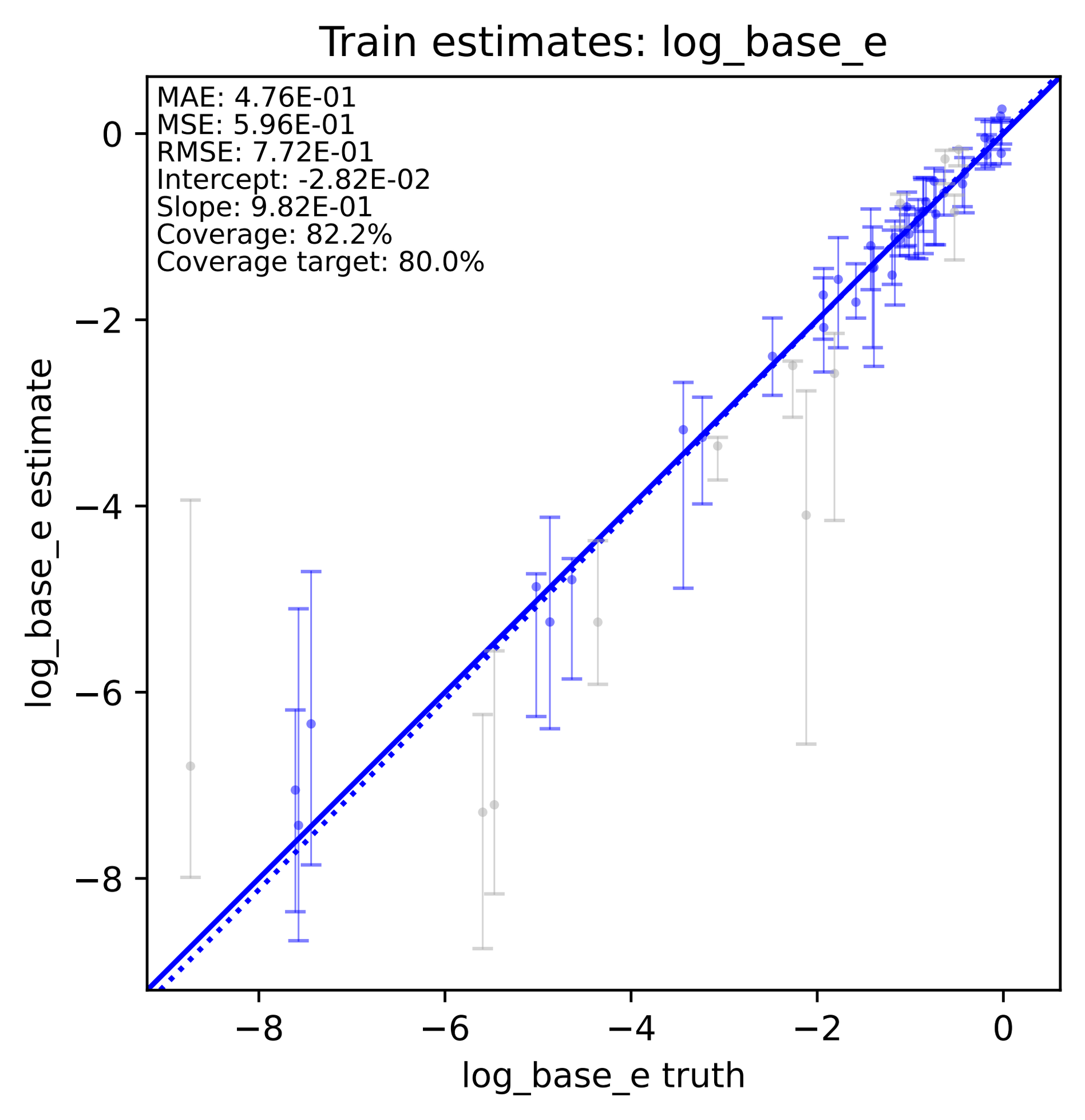

Density: training aux. data
N_bar
sim: 0.96 (0.63, 1.04)
emp: 0.91 (0.91, 0.91)
age_var
sim: 0.90 (-0.45, 2.80)
emp: 0.84 (0.84, 0.84)
brlen_var
sim: 0.32 (-1.06, 2.48)
emp: 0.76 (0.76, 0.76)
colless
sim: -1.41 (-1.66, -0.54)
emp: -0.57 (-0.57, -0.57)
f_dat_0
sim: 3.39 (0.35, 18.91)
emp: 3.95 (3.95, 3.95)
f_dat_1
sim: 2.67 (1.02, 5.62)
emp: 1.05 (1.05, 1.05)
log10_B1
sim: 2.02 (0.83, 2.21)
emp: 1.26 (1.26, 1.26)
log10_age_mean
sim: 0.47 (-0.20, 1.54)
emp: 0.67 (0.67, 0.67)
log10_brlen_mean
sim: 0.18 (-0.49, 1.29)
emp: 0.40 (0.40, 0.40)
log10_root_age
sim: 1.13 (0.46, 1.95)
emp: 1.03 (1.03, 1.03)
log10_tree_length
sim: 2.71 (1.99, 3.61)
emp: 2.27 (2.27, 2.27)
n_dat_0
sim: 6.37 (3.09, 7.88)
emp: 5.01 (5.01, 5.01)
n_dat_1
sim: 6.16 (3.14, 7.07)
emp: 3.69 (3.69, 3.69)
num_taxa
sim: 194.00 (13.00, 298.00)
emp: 38.00 (38.00, 38.00)
prop_taxa
sim: 1.00 (1.00, 1.00)
emp: 1.00 (1.00, 1.00)
treeness
sim: -0.28 (-0.42, -0.20)
emp: -0.83 (-0.83, -0.83)
Density
Variable

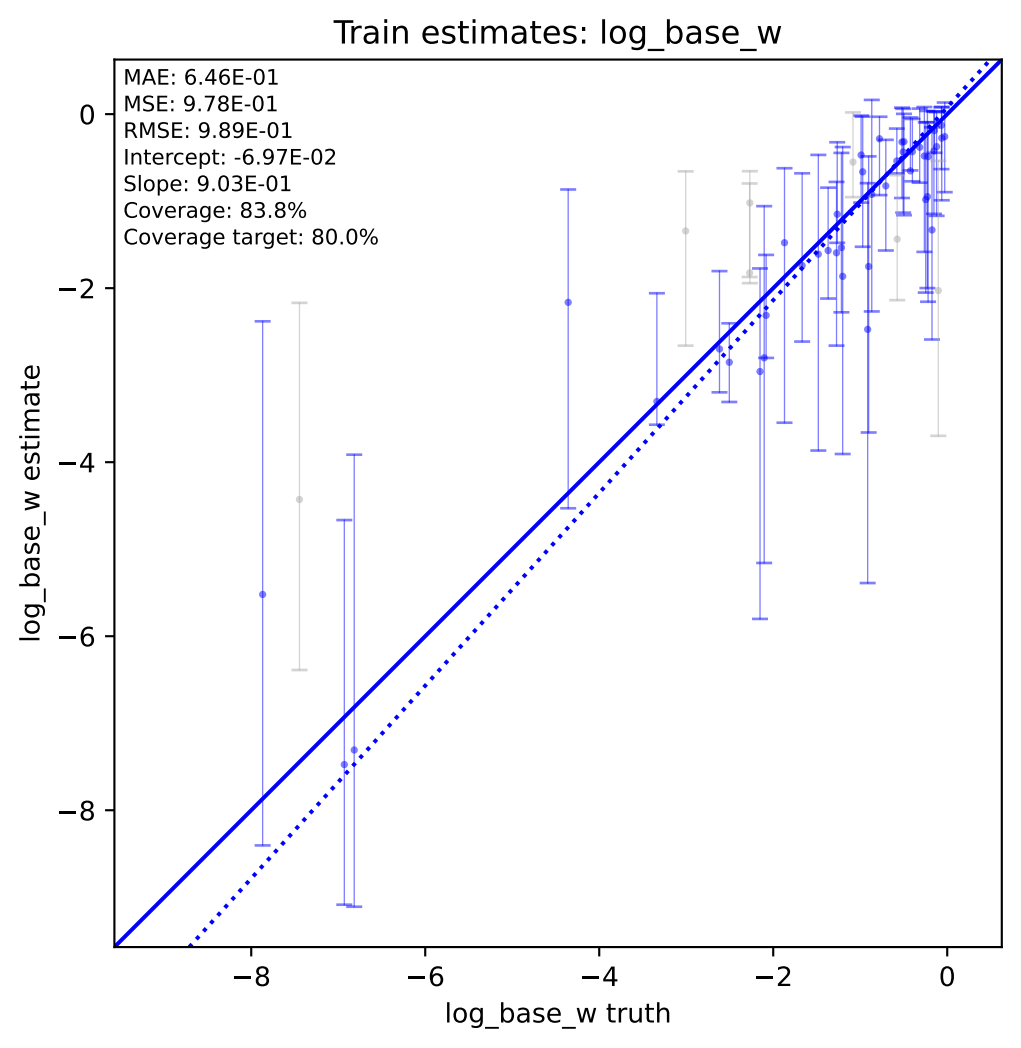

Density: training labels
exp_scalar_w
sim: 0.38 (0.14, 0.95)
emp: 0.22 (0.22, 0.22)
log_base_b
sim: -1.53 (-8.59, -0.05)
emp: -1.75 (-1.75, -1.75)
log_base_d
sim: -1.17 (-8.33, -0.04)
emp: -6.07 (-6.07, -6.07)
log_base_e
sim: -2.47 (-8.88, -0.07)
emp: -6.44 (-6.44, -6.44)
log_base_w
sim: -1.22 (-8.12, -0.04)
emp: -0.13 (-0.13, -0.13)
Density
Variable

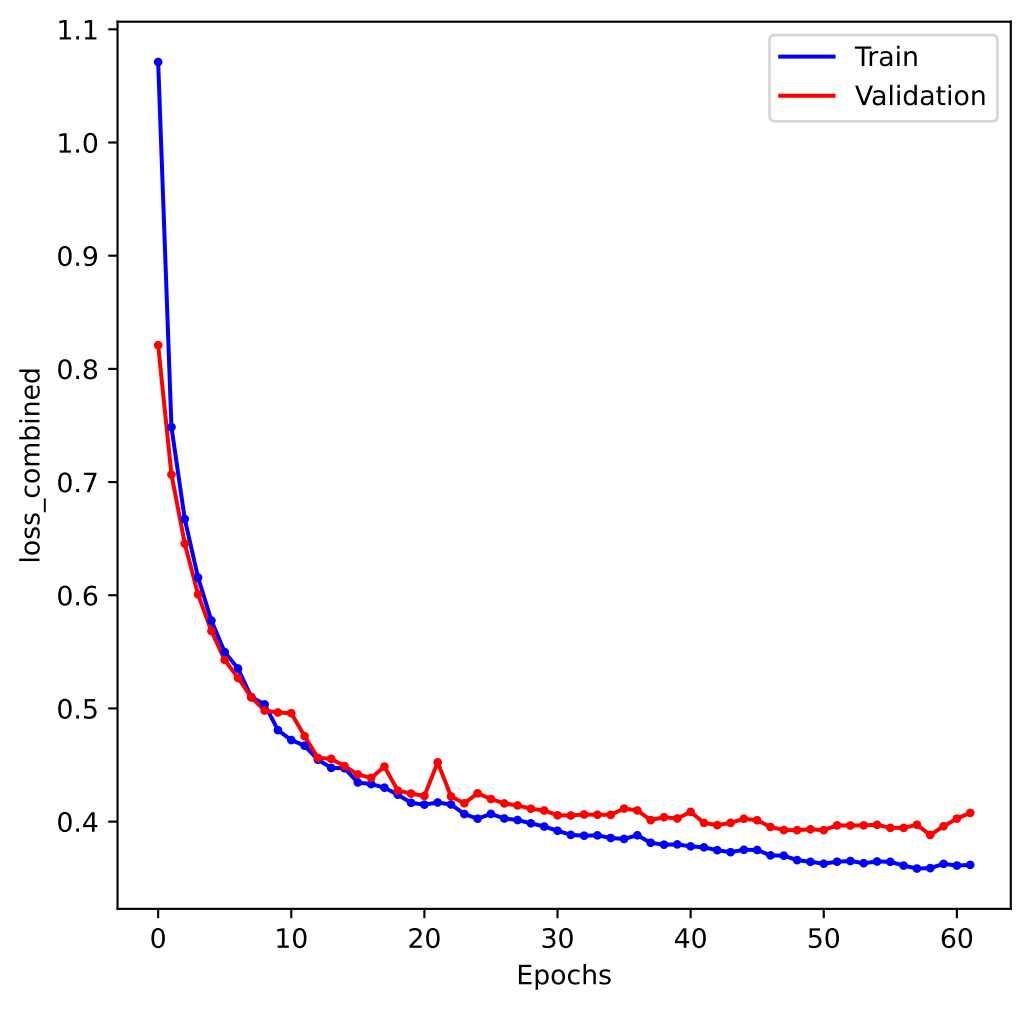

Train estimates: exp_scalar_w
MAE: 1.28E-01
MSE: 3.37E-02
RMSE: 1.84E-01
Intercept: 9.13E-02
Slope: 9.48E-01
Coverage: 82.4%
Coverage target: 80.0%
exp_scalar_w estimate
exp_scalar_w truth
0.2
0.4
0.6
0.8
1.0

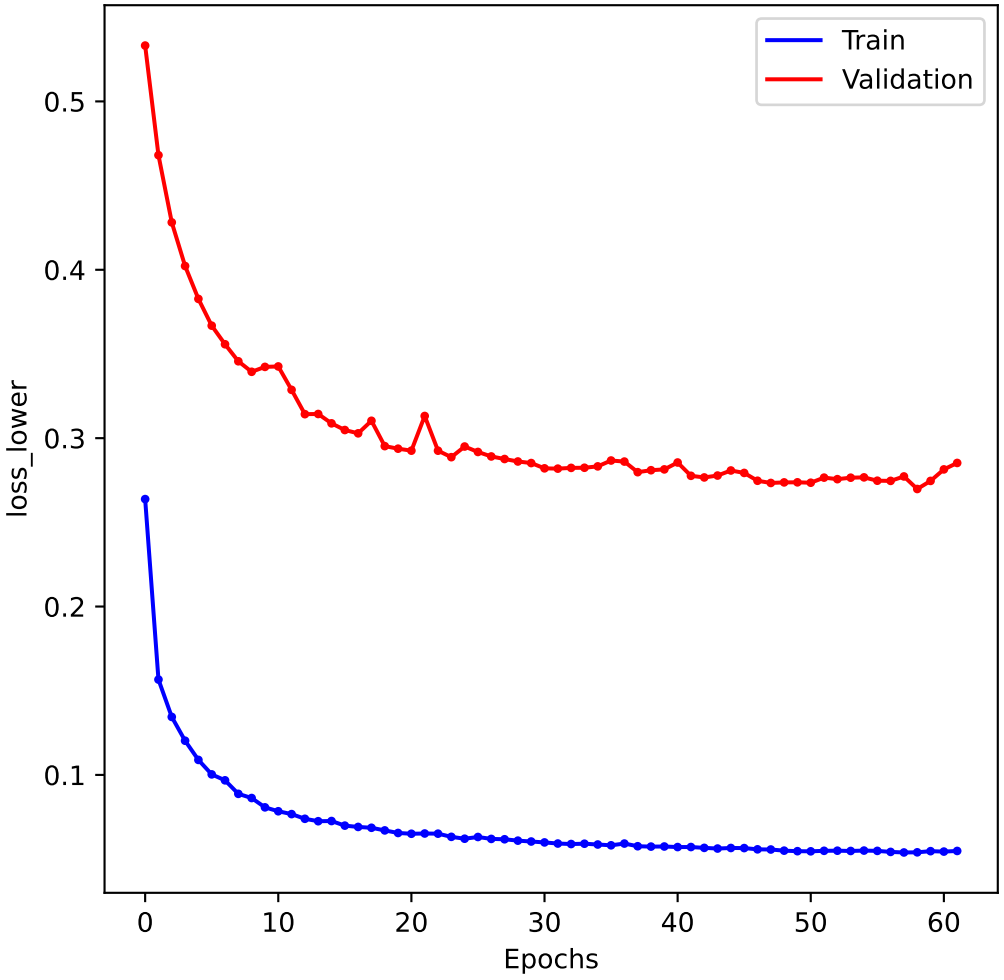

Train estimates: log_base_b
MAE: 6.33E-01
MSE: 1.41E+00
RMSE: 1.19E+00
Intercept: -1.81E-01
Slope: 1.15E+00
Coverage: 79.8%
Coverage target: 80.0%
log_base_b estimate
log_base_b truth
0
−2
−4
−6
−8
−10
−10
−8
−6
−4
−2
0

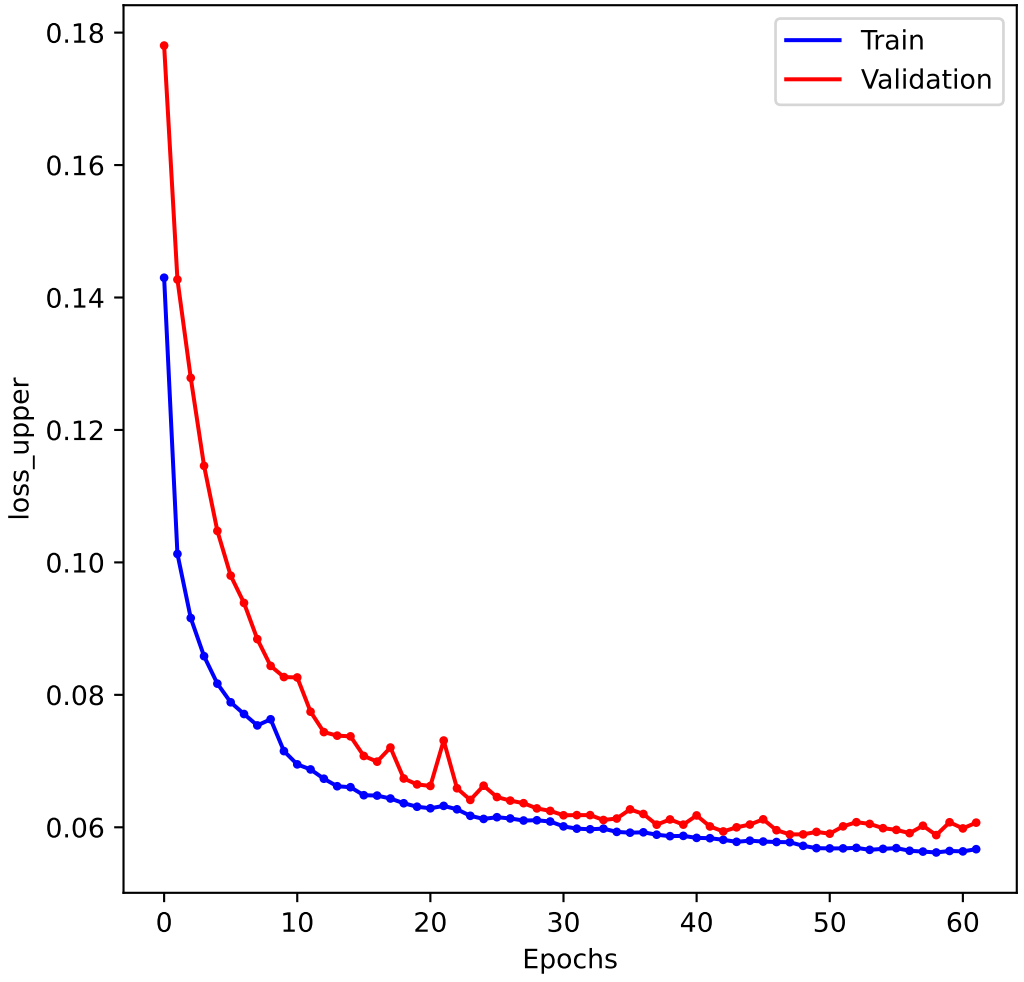

Train estimates: log_base_d
MAE: 1.66E-01
MSE: 8.67E-02
RMSE: 2.94E-01
Intercept: 5.14E-02
Slope: 1.03E+00
Coverage: 84.5%
Coverage target: 80.0%
log_base_d estimate
log_base_d truth
0
−2
−4
−6
−8
−8
−6
−4
−2
0

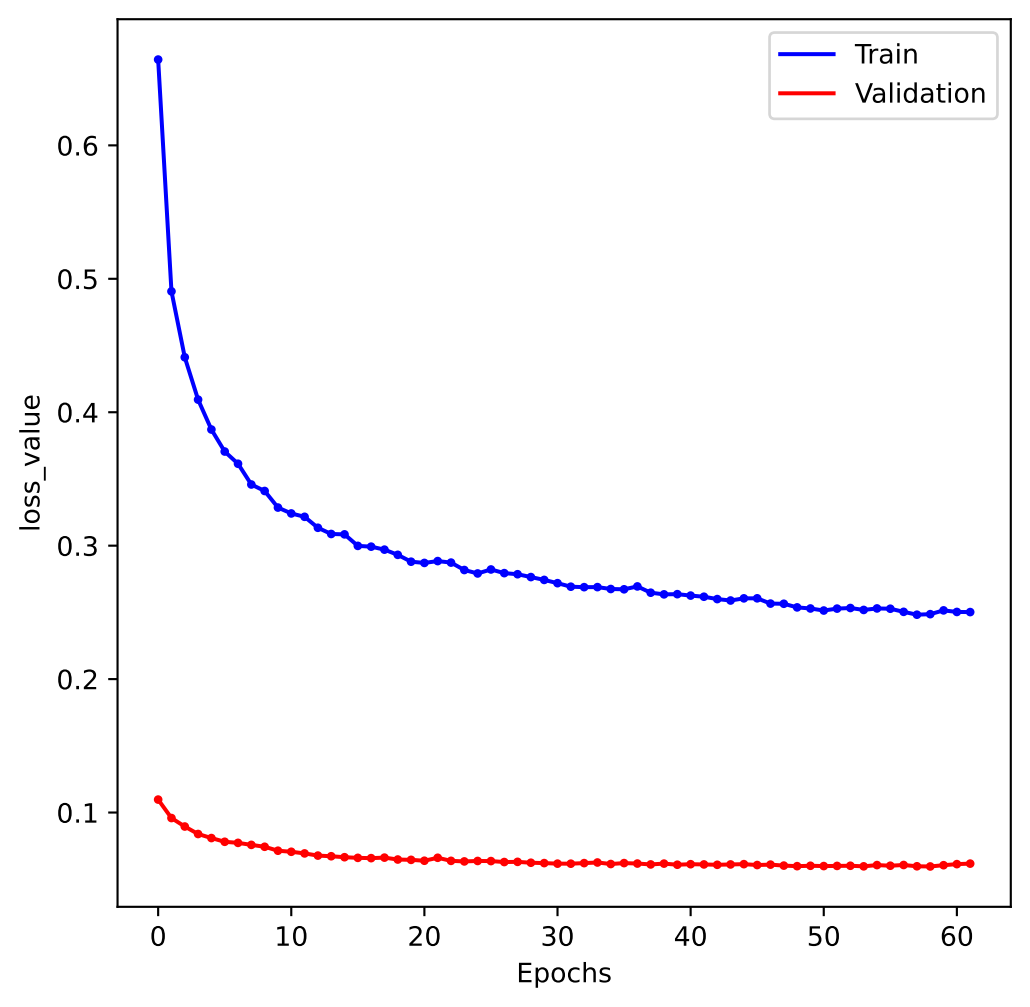

Train estimates: log_base_e
MAE: 4.76E-01
MSE: 5.96E-01
RMSE: 7.72E-01
Intercept: -2.82E-02
Slope: 9.82E-01
Coverage: 82.2%
Coverage target: 80.0%
log_base_e estimate
log_base_e truth
0
−2
−4
−6
−8
−8
−6
−4
−2
0

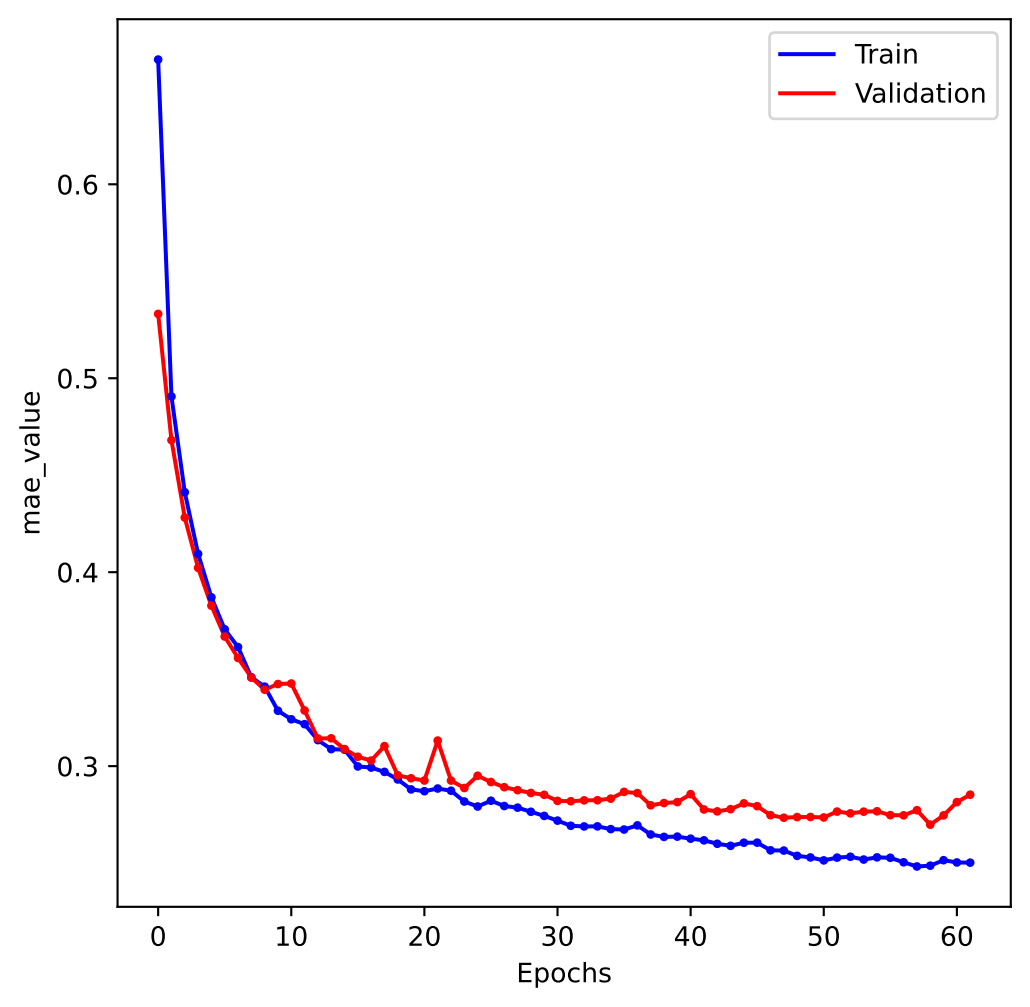

Train estimates: log_base_w
MAE: 6.46E-01
MSE: 9.78E-01
RMSE: 9.89E-01
Intercept: -6.97E-02
Slope: 9.03E-01
Coverage: 83.8%
Coverage target: 80.0%
log_base_w estimate
log_base_w truth
0
−2
−4
−6
−8
−8
−6
−4
−2
0

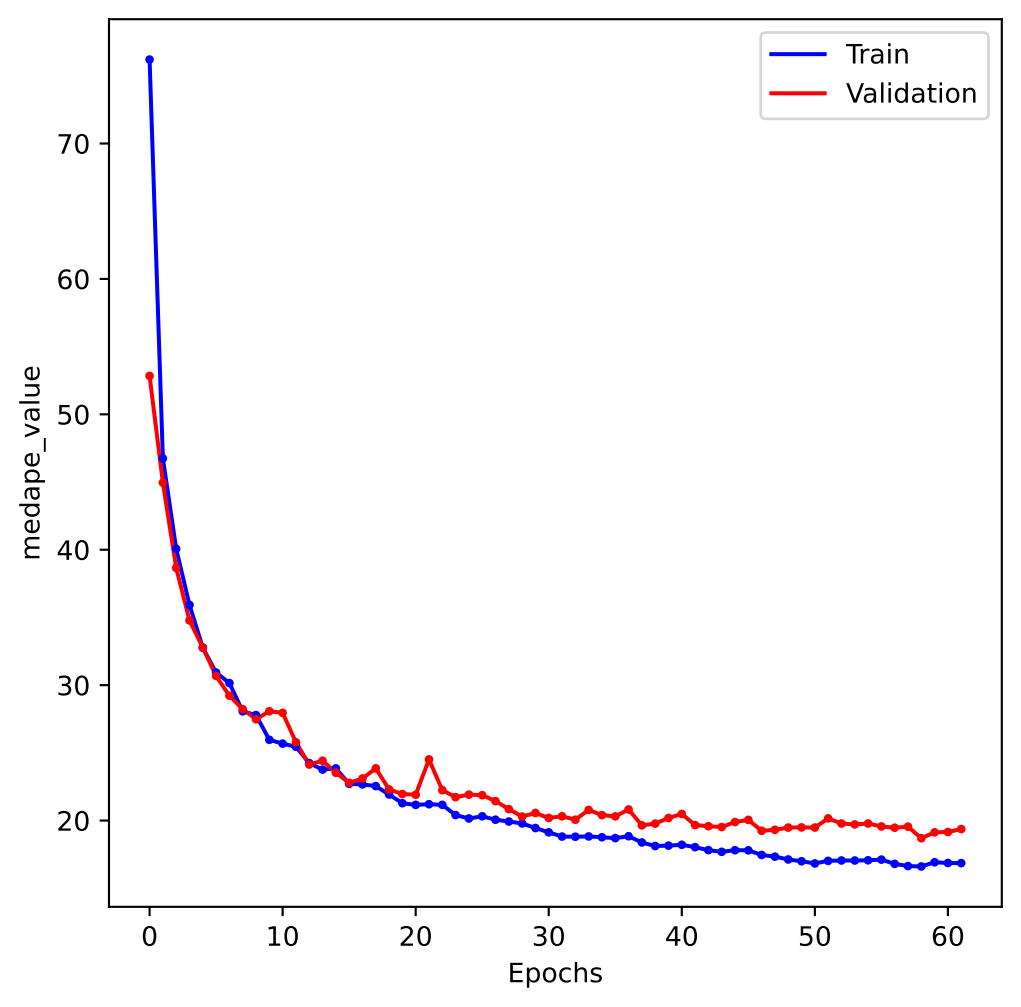

Train
Validation
loss_combined
Epochs
1.1
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0
10
20
30
40
50
60

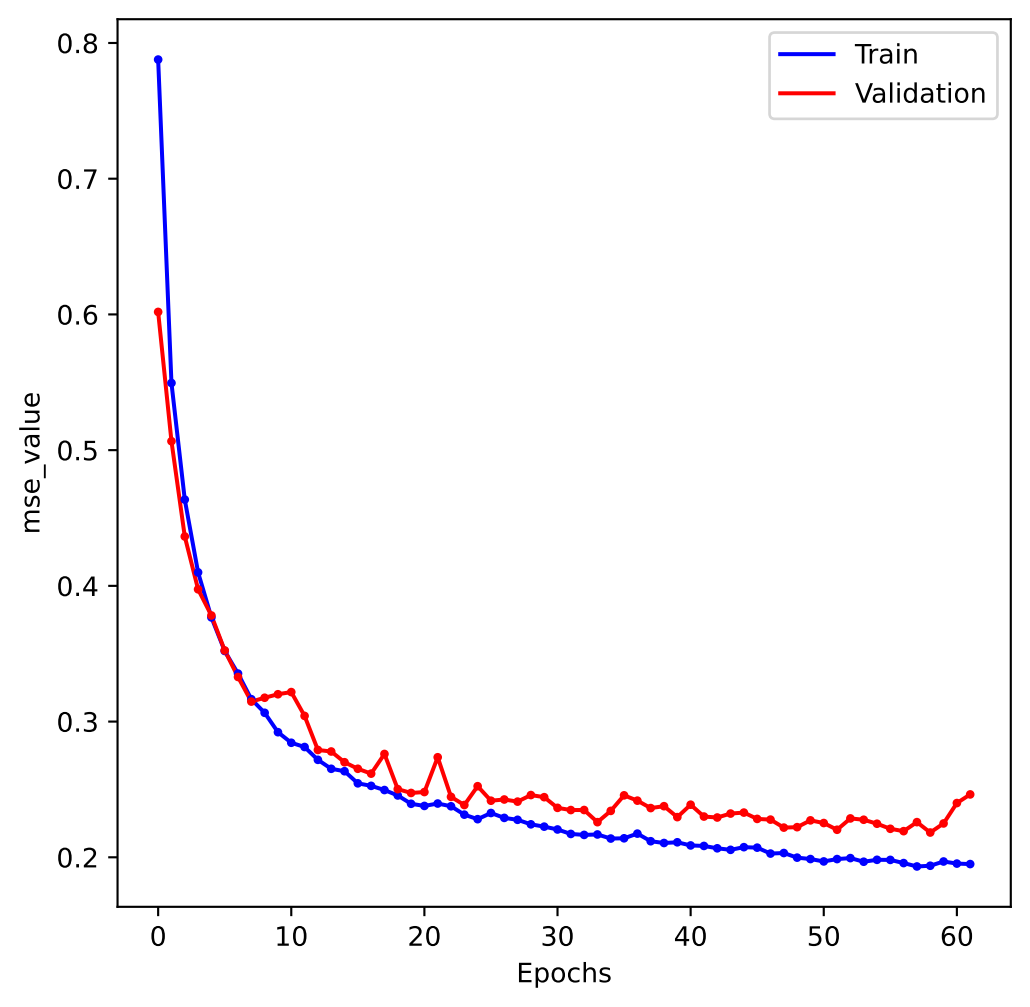

Train
Validation
loss_lower
0.5
0.4
0.3
0.2
0.1
0
10
20
30
40
50
60
Epochs

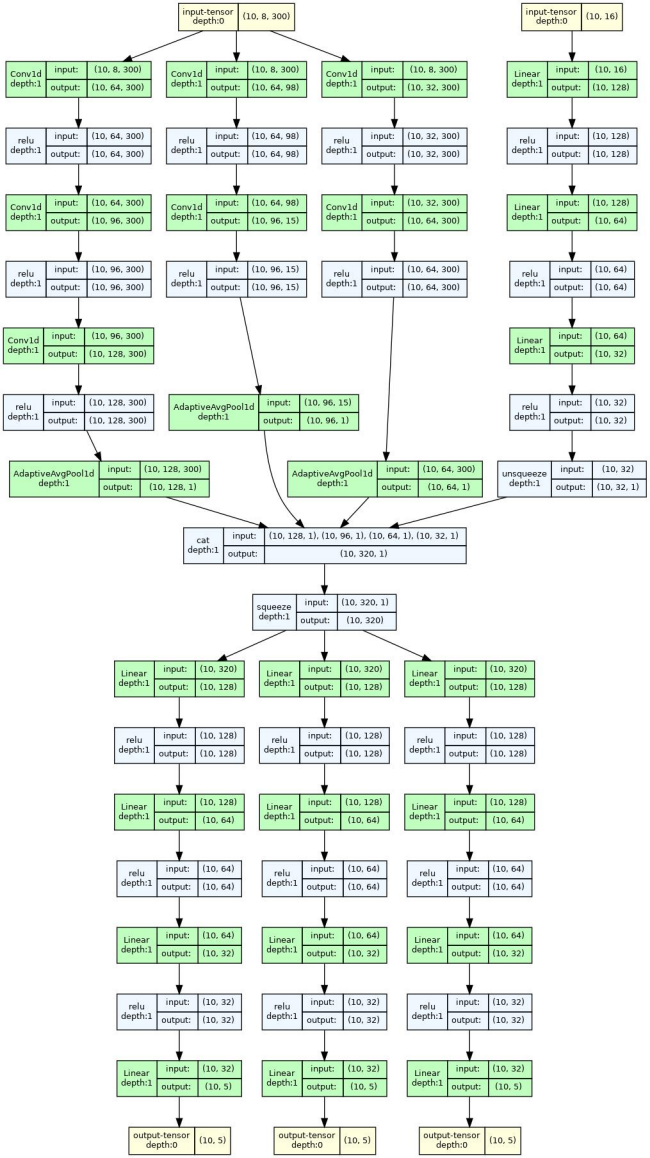

Train
Validation
loss_upper
0.18
0.16
0.14
0.12
0.10
0.08
0.06
0
10
20
30
40
50
60
Epochs


Train
Validation
loss_value
0.6
0.5
0.4
0.3
0.2
0.1
0
10
20
30
40
50
60
Epochs


Train
Validation
mae_value
0.6
0.5
0.4
0.3
0
10
20
30
40
50
60
Epochs


Train
Validation
medape_value
70
60
50
40
30
20
0
10
20
30
40
50
60
Epochs


Train
Validation
mse_value
Epochs
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0
10
20
30
40
50
60


input-tensor depth:0
(10, 8, 300)
input-tensor depth:0
(10, 16)
Conv1d depth:1
input: (10, 8, 300)
output: (10, 64, 300)
Conv1d depth:1
input: (10, 8, 300)
output: (10, 64, 98)
Conv1d depth:1
input: (10, 8, 300)
output: (10, 32, 300)
Linear depth:1
input: (10, 16)
output: (10, 128)
relu depth:1
input: (10, 64, 300)
output: (10, 64, 300)
relu depth:1
input: (10, 64, 98)
output: (10, 64, 98)
relu depth:1
input: (10, 32, 300)
output: (10, 32, 300)
relu depth:1
input: (10, 128)
output: (10, 128)
Conv1d depth:1
input: (10, 64, 300)
output: (10, 96, 300)
Conv1d depth:1
input: (10, 64, 98)
output: (10, 96, 15)
Conv1d depth:1
input: (10, 32, 300)
output: (10, 64, 300)
Linear depth:1
input: (10, 128)
output: (10, 64)
relu depth:1
input: (10, 96, 300)
output: (10, 96, 300)
relu depth:1
input: (10, 96, 15)
output: (10, 96, 15)
relu depth:1
input: (10, 64, 300)
output: (10, 64, 300)
relu depth:1
input: (10, 64)
output: (10, 64)
Conv1d depth:1
input: (10, 96, 300)
output: (10, 128, 300)
Linear depth:1
input: (10, 64)
output: (10, 32)
relu depth:1
input: (10, 128, 300)
output: (10, 128, 300)
AdaptiveAvgPool1d depth:1
input: (10, 96, 15)
output: (10, 96, 1)
relu depth:1
input: (10, 32)
output: (10, 32)
AdaptiveAvgPool1d depth:1
input: (10, 128, 300)
output: (10, 128, 1)
AdaptiveAvgPool1d depth:1
input: (10, 64, 300)
output: (10, 64, 1)
unsqueeze depth:1
input: (10, 32)
output: (10, 32, 1)
cat depth:1
input: (10, 128, 1), (10, 96, 1), (10, 64, 1), (10, 32, 1)
output: (10, 320, 1)
squeeze depth:1
input: (10, 320, 1)
output: (10, 320)
Linear depth:1
input: (10, 320)
output: (10, 128)
Linear depth:1
input: (10, 320)
output: (10, 128)
Linear depth:1
input: (10, 320)
output: (10, 128)
relu depth:1
input: (10, 128)
output: (10, 128)
relu depth:1
input: (10, 128)
output: (10, 128)
relu depth:1
input: (10, 128)
output: (10, 128)
Linear depth:1
input: (10, 128)
output: (10, 64)
Linear depth:1
input: (10, 128)
output: (10, 64)
Linear depth:1
input: (10, 128)
output: (10, 64)
relu depth:1
input: (10, 64)
output: (10, 64)
relu depth:1
input: (10, 64)
output: (10, 64)
relu depth:1
input: (10, 64)
output: (10, 64)
Linear depth:1
input: (10, 64)
output: (10, 32)
Linear depth:1
input: (10, 64)
output: (10, 32)
Linear depth:1
input: (10, 64)
output: (10, 32)
relu depth:1
input: (10, 32)
output: (10, 32)
relu depth:1
input: (10, 32)
output: (10, 32)
relu depth:1
input: (10, 32)
output: (10, 32)
Linear depth:1
input: (10, 32)
output: (10, 5)
Linear depth:1
input: (10, 32)
output: (10, 5)
Linear depth:1
input: (10, 32)
output: (10, 5)
output-tensor depth:0
(10, 5)
output-tensor depth:0
(10, 5)
output-tensor depth:0
(10, 5)